PAPER TITLE

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ABSTRACT

Plasma in a Tokamak fusion reactor can often experience disruption events, causing damage to the interior reactor walls and resulting in a current drop to zero. The existence of a computationally inexpensive model for identifying disruptive shots and accurately identifying the time of disruption can assist the scientific community by efficiently auto-labeling experimental data for further analysis. We took an existing labeled disruption dataset from the DIII-D Tokamak reactor, utilizing shot numbers 125500-186000 (40,000 total) to develop a new disruption labeling model that can generate automated labeling up to the current day. We train a CNN (Convolutional Neural Network) to perform boolean disruption/not-disruption prediction and achieve a 99% recall and 91% precision on our test dataset. We additionally develop a convolutional heuristic that accurately identifies time of disruption with a standard deviation $\sigma = 8.7\,\mathrm{m\,sec}$.

1 Introduction

State the problem, motivation, and contributions.

2 Methods

Describe data, models, training, and evaluation.

3 Results

Report quantitative and qualitative findings. Reference figures and tables as needed.

4 Discussion

Interpret results, note limitations, and compare to prior work.

5 Conclusion

Summarize findings and outline future work.

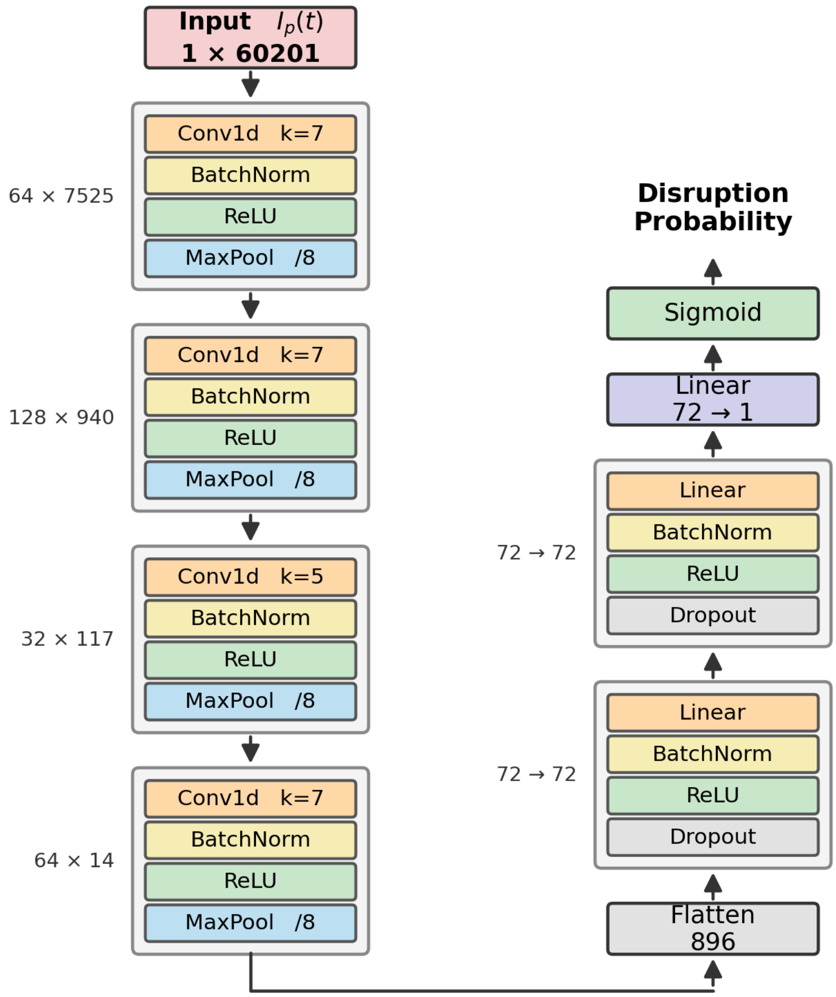

Figure 1: Figure caption.

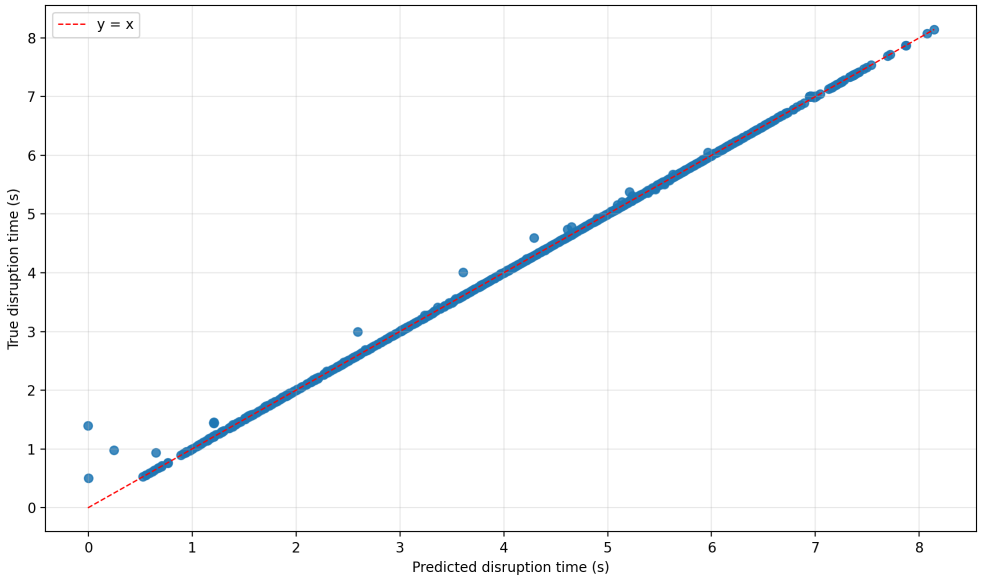

Figure 2: Figure caption.

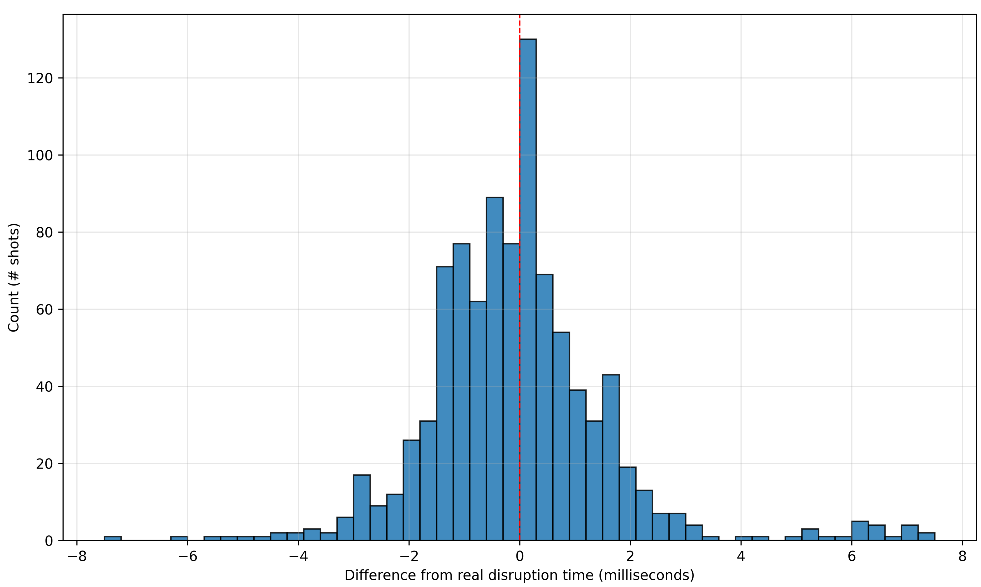

Figure 3: Figure caption.